ACRME Hard Constraints Reference

Classification: Principal Cloud Architect — Placement Engine Design
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Status: Production Ready

> **v2.2 reconciliation note (2 Sep 2026).** Under Requirements Baseline v2.2 and ADR-002 v2.2 the **single governed quota pool** is the **primary** model: one pool per region/quota family covers Prod + NonProd/CVAL + DR, with Prod and DR protected by **logical earmarks** (`Prod_Reserved_Floor`, `DR_Earmark_vCPU`) rather than physical group separation. The **Two-Group Quota Architecture** referenced by HC-3 and HC-7 below is retained as the sanctioned **exception topology** (used only when Azure Quota Group limits or a mandatory Prod-isolation boundary make one pool impossible). The HC-3/HC-7 arithmetic is unchanged and remains correct; in the single-pool model the terms map as `Effective_NonProd_Ceiling → Allocatable_NonProd`, `NonProd_DR_Group_Limit → Pool_Limit`, and `DR_Floor_vCPU → DR_Earmark_vCPU` (max-not-sum, DR-017). Also: the EU cross-geo DR extension region is **Switzerland North** (REG-002). See ADR-002 v2.2, the Calculation Logic Reference v2.2 (Scenario 8/9), the FDD §4.2, and the TDD §8.3.
>
> ⚠️ **Middle East DR — current legal position is `DR_NOT_OFFERED` (DEC-001, under legal review).** Baseline v2.2 records that **Legal has taken ownership of the Middle East programme** and that, because a large share of Middle East customers are government/medical-associated, **data-sovereignty / data-residency laws mean cross-border DR cannot meet residency requirements — so DR is currently NOT offered in the Middle East** (baseline §2 Strategic Drivers, §5.2 Out of Scope, §6 Region Strategy, **DR-014**). This is a **pending legal/business decision (DEC-001)**, listed among the programme's remaining major architectural risks (baseline §25). Consequences for placement below:
> - **The current default for a Middle East geography is `dr_region = NOT_OFFERED`.** The engine records the seed with no DR region and **does not** auto-assign any cross-geo DR. Middle East **production may still exist** without DR (DR-014).
> - **Switzerland North is a pre-configured, conditional cross-geo extension path only.** It is held in `PlacementPolicy` so that DR can be turned on quickly **if and only if Legal explicitly approves Middle East DR via DEC-001**. Until that approval is recorded, the extension path is **inactive** and the HC-10 canonical example below is a conditional design, not a live placement.
> - The HC-10 Cross-Geo Extension mechanics and VR-6 below therefore describe the **approved-path behaviour that applies only after DEC-001 clears**; the **default legal state overrides them with `DR_NOT_OFFERED`**. See ADR-001, ADR-005, the FDD §4.4/§8, the TDD §2.2/§9, and the Calculation Logic Reference Scenario 4.

Overview

This document consolidates all **Hard Constraints (HC-1 through HC-10)** governing regional placement decisions in the **Azure Capacity Reservation Management Engine (ACRME)**. Hard constraints are **binary pass/fail gates** — any region failing a hard constraint is excluded from scoring entirely, not penalized with a lower score.

Enforcement Architecture

Hard constraints are enforced at two stages in the placement decision pipeline:

Stage 1 — Pre-filtering (before candidate set formation):
- **HC-8** GEOGRAPHY_CONTAINMENT
- **HC-9** STANDARD_REGION_ONLY

Stage 2 — Hard Constraint Gate (after geography/class filtering, before scoring):
- **HC-1** REGION_SEPARATION
- **HC-2** CAPACITY_FLOOR
- **HC-3** QUOTA_FLOOR
- **HC-4** DR_SEPARATION_CLASS
- **HC-5** ZONE_AVAILABILITY
- **HC-6** DR_COVERAGE_FLOOR
- **HC-7** DR_FLOOR_INTEGRITY
- **HC-10** CROSS_GEO_EXTENSION_PATH_APPROVED

Regions surviving both stages enter the **scoring pipeline** where soft objectives (`PS_Prod` , `PS_NonProd` , `PS_DR`) rank them.

Source Documents

- **HC-1 through HC-7:** `multi_region_placement_design.md` §27 (Hard Constraints)
- **HC-8, HC-9, HC-10:** `acrme_production_readiness_review_and_architecture.md` §27 (Production-Ready Region Selection)
- **Design rationale:** `design_change_summary.md` (Decisions D1–D9)

Part 1 — Hard Constraint Definitions

HC-1: REGION_SEPARATION [UPDATED]

Category: Isolation
Source: Multi-Region Placement Design §27; Decision D8
Status: Updated in Pass 2 — NonProd/DR co-location now permitted

Definition

```
NonProd_region ≠ Prod_region
DR_region      ≠ Prod_region
[Constraint DR_region ≠ NonProd_region REMOVED — NonProd and DR may share a region]
```

Rationale

NonProd/DR co-location is permitted to allow **DR overflow capacity reuse** from the NonProd CRG. This change (Decision D8) enables the engine to leverage unused NonProd capacity as a DR buffer, reducing the minimum region count from 4 to 3.

With 3 regions:
Prod is isolated; NonProd and DR both draw from the remaining 2 regions (they may land on the same region or on different ones — determined by HC-6 and PS score).

With 4 regions:
Prod eliminates 1; NonProd selects from 3; DR may share with NonProd or use the remaining regions — whichever satisfies HC-6 and maximises PS_DR.

Implementation

- Prod region is excluded from NonProd candidate set
- Prod region is excluded from DR candidate set
- NonProd region is **not** excluded from DR candidate set (co-location allowed)
- CVAL (alias for NonProd) and Prod must not share a region
- CVAL and DR may share a region only under the approved policy (HC-1 update per D8)

Validation Rules (VR)

- **VR-1:** Prod and DR must not share a region
- **VR-2:** CVAL and Prod must not share a region
- **VR-3:** CVAL and DR may share a region only under the approved policy

Evidence Tag

`[Derived]` from Azure Well-Architected Framework (Reliability pillar)

HC-2: CAPACITY_FLOOR

Category: Capacity
Source: Multi-Region Placement Design §27
Status: Stable

Definition

```
CR headroom in candidate region ≥ minimum_reservation_units

where:
  minimum_reservation_units = 2 × requested_vm_count × SKU_vCPU_count  (default multi-
plier: 2)

  CR headroom = total_free_slots (in the target CRG type: Prod/NonProd/DR)
```

Rationale

Ensures sufficient capacity buffer exists in the region before placement. The **2× multiplier** provides headroom for growth and prevents placement into near-exhaustion regions.

Implementation

- Read `RegionalSnapshot.prod_crg_free_slots`, `nonprod_crg_free_slots`, or `dr_crg_free_slots` (per environment type)
- Compare against `minimum_reservation_units` (configurable per PlacementPolicy)
- **Regions at capacity floor are excluded from scoring entirely**

Configuration

- **Default multiplier:** 2
- **Configurable via:** `PlacementPolicy.capacity_floor_multiplier`
- **Overridable per geography or SKU class**

Evidence Tag

`[Undocumented – architectural judgement]`

HC-3: QUOTA_FLOOR [UPDATED]

Category: Quota
Source: Multi-Region Placement Design §27; Design Change Summary QG-3
Status: Updated in Pass 2 — now reads from Quota Groups, not per-subscription QuotaRecord

Definition

```
For Prod placement in region R:
  REJECT if: Prod_Group_headroom(R) < requested_vm_count × vCPU_per_instance

For NonProd placement in region R:
  REJECT if: nonprod_headroom(R) < requested_vm_count × vCPU_per_instance
             (nonprod_headroom uses effective_nonprod_ceiling – floor-adjusted, not
raw group_limit)

For DR placement in region R:
  REJECT if: dr_headroom(R) < target_dr_qty × vCPU_per_instance
             (target_dr_qty = customer_prod_vm_count × dr_ratio)
```

Data Source

- **Primary:** `QuotaGroup` entity (Cosmos DB), synced to `RegionalSnapshot` (Redis)
- **Fields:**
- `Prod_Group_headroom(R)` = Prod group available quota in region R
- `nonprod_headroom(R)` = NonProd+DR group available quota **minus DR floor** (see HC-7)
- `dr_headroom(R)` = DR portion of NonProd+DR group quota
- **Legacy field:** `QuotaRecord.deployment_headroom` is superseded for placement checks; retained for per-subscription quota monitoring only

Formulas

Prod Group Headroom:

```
Prod_Group_headroom(R) = Prod_Group_Limit(R) - Prod_Group_Used(R)

where:
  Prod_Group_Limit(R) = base_subscription_quota_limit × (1 + emer-
gency_transfer_headroom_vcpu / base_limit)
  Prod_Group_Used(R)  = Σ (prod_crg.quantity × vCPU) for all Prod CRGs in region R
```

NonProd Headroom (effective ceiling):

```
nonprod_headroom(R) = effective_nonprod_ceiling(R) - nonprod_quota_used(R)

where:
  effective_nonprod_ceiling(R) = NonProd_DR_Group_Limit(R) - DR_Floor_vCPU(R)
  NonProd_DR_Group_Limit(R)    = base_subscription_quota_limit × (1 + emer-
gency_transfer_headroom_vcpu / base_limit)
  DR_Floor_vCPU(R)              = potential_dr_demand(R) × vCPU × dr_ratio_max
  nonprod_quota_used(R)         = Σ (nonprod_crg.quantity × vCPU) for all NonProd CRGs
in region R
```

DR Headroom:

```
dr_headroom(R) = DR_Floor_vCPU(R) - dr_quota_used(R)

where:
  dr_quota_used(R) = Σ (dr_crg.quantity × vCPU) for all DR CRGs in region R
```

Validation Sequence

1. **HC-3 (Quota Floor):** Check raw headroom availability
2. **HC-7 (DR Floor Integrity):** Check floor compliance (applies only to NonProd placements)

POC Blocker

B-1: Azure Quota Groups functionality must be GA-available in target tenant/region. If `Microsoft.Capacity/resourceProviders/.../groupQuotas` returns 404, escalate to Azure Support. **POC-30** gates this.

Evidence Tag

`[Documented]` — Azure Quota Groups Preview API documentation

HC-4: DR_SEPARATION_CLASS

Category: Resilience
Source: Multi-Region Placement Design §27
Status: Stable (non-paired regions only)

Definition

```
DR_region must have Separation_Class(DR_region, Prod_region) = HIGH
```

Rationale

Ensures non-correlated failure domains between Prod and DR regions. This constraint applies **only to non-paired regions** (e.g., standalone configuration with 3-4 explicitly chosen regions). Paired regions (e.g., East US / West US) are assumed to have HIGH separation by Azure design.

Separation Class Matrix

Prod Region	DR Region	Separation Class	Eligible?
East US	West US	HIGH	✅ Yes (paired)
East US	Central US	MEDIUM	❌ No
East US	West Europe	HIGH	✅ Yes (cross-geo)
UAE North	Saudi Arabia Central	MEDIUM	❌ No (same geography)
UAE North	Switzerland North	HIGH	⚠️ Separation-eligible, but **inactive** — Middle East DR is `DR_NOT_OFFERED` pending DEC-001 (DR-014); usable only after legal approval

Implementation

- Read `SeparationClassRegistry` (static config or Azure metadata service)
- For non-paired regions, validate `Separation_Class(DR, Prod) = HIGH`
- Paired regions bypass this check (assumed HIGH by default)

Evidence Tag

`[Undocumented — architectural judgement]`

HC-5: ZONE_AVAILABILITY

Category: Resilience
Source: Multi-Region Placement Design §27
Status: Stable

Definition

```
Candidate region must have ≥ 2 physical Availability Zones
```

Rationale

Single-zone regions are excluded for all non-dev environments. Multi-zone deployment is a **Well-Architected Framework Reliability pillar** requirement.

Implementation

- Read `RegionalSnapshot.zone_count` (populated from Azure metadata)
- Reject regions where `zone_count < 2`
- **Applies to:** Prod, CVAL, DR placements

- **Exemption:** Dev environments may bypass this constraint (configurable via `Placement-Policy.allow_single_zone_dev`)

Examples

Region	Zone Count	Eligible?
East US	3	✅ Yes
West Europe	3	✅ Yes
Central India	3	✅ Yes
Brazil South	1	❌ No
West US 3	3	✅ Yes

Evidence Tag

`[Undocumented – architectural judgement]`

HC-6: DR_COVERAGE_FLOOR [NEW]

Category: DR Capacity
Source: Multi-Region Placement Design §27; Design Change Summary CR/CRG-1; Decision D8
Status: New in Pass 2 — operationalises NonProd/DR co-location

Definition

```
REJECT DR placement in region R if combined DR + NonProd overflow capacity
is insufficient to absorb the new customer's DR demand:

  dr_crg_free_slots(R) + nonprod_crg_effective_free(R) < customer_requested_dr_slots

where:
  customer_requested_dr_slots = prod_vm_count × dr_ratio_max
  nonprod_crg_effective_free  = nonprod_crg_free_slots - non-
prod_crg_dr_overflow_reserve

  dr_ratio_max = 0.40  (policy constant – upper bound for DR-to-Prod ratio)
```

Rationale

Since DR and NonProd may share a region (HC-1 constraint removed per D8), the engine must verify that the **combined capacity pool** can absorb the customer's DR need — either from the DR CRG directly, or from the NonProd CRG's reserved overflow headroom.

This constraint operationalises HC-1 co-location at the capacity selection level.

Data Requirements

Per-CRG-type snapshot fields (added in Pass 2):
- `nonprod_crg_free_slots(R)` — available NonProd CR capacity
- `nonprod_crg_effective_free(R)` — NonProd capacity minus DR overflow reserve

- `nonprod_crg_dr_overflow_reserve(R)` — capacity earmarked for DR failover absorption
- `dr_crg_free_slots(R)` — available DR CR capacity

Implementation Pseudocode

```
def validate_HC6_DR_coverage_floor(region, customer_prod_vm_count):
    snapshot = get_regional_snapshot(region)

    customer_requested_dr_slots = customer_prod_vm_count * DR_RATIO_MAX  # 0.40

    combined_dr_capacity = (snapshot.dr_crg_free_slots +
                            snapshot.nonprod_crg_effective_free)

    if combined_dr_capacity < customer_requested_dr_slots:
        return REJECT, "HC-6 DR_COVERAGE_FLOOR: Insufficient combined DR+NonProd
overflow capacity"

    return PASS
```

Policy Constants

- **dr_ratio_min:** 0.30 (minimum DR-to-Prod ratio)
- **dr_ratio_max:** 0.40 (maximum DR-to-Prod ratio, used in HC-6 floor calculation)
- **dr_ratio_target:** 0.35 (recommended ratio)

Evidence Tag

`[Derived]` from Decision D8 (NonProd/DR co-location)

HC-7: DR_FLOOR_INTEGRITY [NEW]

Category: Quota Protection
Source: Multi-Region Placement Design §27; Design Change Summary QG-4
Status: New in Pass 2 — enforces DR floor within Two-Group Quota Architecture

Definition

```
REJECT NonProd placement in region R if placing the requested quantity
would push nonprod_quota_used above effective_nonprod_ceiling:

  nonprod_quota_used(R) + (requested_vm_count × vCPU_per_instance) > effect-
ive_nonprod_ceiling(R)

where:
  effective_nonprod_ceiling(R) = NonProd_DR_Group_Limit(R) - DR_Floor_vCPU(R)
```

Rationale

Prevents NonProd allocation from **encroaching on the protected DR quota floor**. Within the Two-Group Quota Architecture, the NonProd+DR group has a shared limit; the DR floor reserves a portion for DR use only.

Evaluation Order

1. **HC-3 (Quota Floor):** Check raw headroom (`nonprod_headroom ≥ requested`)
2. **HC-7 (DR Floor Integrity):** Check floor compliance (`used + requested ≤ effective_ceiling`)

HC-3 checks availability; HC-7 checks compliance.

Formulas

Effective NonProd Ceiling:

```
effective_nonprod_ceiling(R) = NonProd_DR_Group_Limit(R) - DR_Floor_vCPU(R)
```

DR Floor (per region):

```
DR_Floor_vCPU(R) = potential_dr_demand(R) × vCPU_per_instance × dr_ratio_max

where:
  potential_dr_demand(R) = Σ prod_crg.quantity for all Prod CRGs that could fail over
to region R
  dr_ratio_max           = 0.40
```

POC Example (GP-06 Topology)

Inputs:
- Prod demand: 80 VMs × 2 vCPU = 160 vCPU
- Base subscription quota limit: 200 vCPU

Group Limits:
- **Prod Group:** 200 × 1.28 = 256 vCPU (28% emergency headroom)
- **NonProd+DR Group:** 200 × 1.28 = 256 vCPU

DR Floor Calculation:

```
DR_Floor_vCPU = 80 VMs × 2 vCPU × 0.40 = 64 vCPU
```

Effective NonProd Ceiling:

```
effective_nonprod_ceiling = 256 - 64 = 192 vCPU
```

Enforcement:

```
If nonprod_quota_used = 150 vCPU:
  New NonProd request for 60 vCPU × 2 = 120 vCPU
  150 + 120 = 270 > 192  ❌ REJECT (HC-7 violation)

If nonprod_quota_used = 100 vCPU:
  New NonProd request for 40 vCPU × 2 = 80 vCPU
  100 + 80 = 180 ≤ 192  ✅ PASS
```

POC Blocker

B-2: Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim) must be validated. **POC-31** measures release latency; confirm < 5 min for Tier RTO compliance.

Evidence Tag

`[Derived]` from Two-Group Quota Architecture design

HC-8: GEOGRAPHY_CONTAINMENT [IMPLIED]

Category: Geography
Source: Production Readiness Review §27
Status: Implied constraint (no standalone definition block)

Definition

```
In Scenario 1 (geography-based placement), derived Prod region must be within
the Standard Capacity Regions for the customer's chosen geography.
```

Rationale

When a customer supplies an Azure geography (e.g., "US", "Europe", "Middle East") rather than a specific region, the engine derives the Prod anchor via `argmax(PS_Prod)` over Standard Capacity Regions **within that geography only**. This constraint ensures the derived Prod region respects the customer's geographic preference.

Enforcement Stage

Stage 1 — Pre-filtering (before candidate set formation)

Validation Rules (VR)

- **VR-4:** Derived Prod region must be within the Standard Capacity Regions for the customer's chosen geography

Implementation

```python
def get_prod_candidates(geography):
    all_regions = get_all_azure_regions()

    # Stage 1: Geography containment (HC-8)
    geo_filtered = [r for r in all_regions
                    if r.geography == geography and
                       r.region_class == "Standard"]

    # Stage 2: Apply HC-1..HC-7, HC-10
    hc_filtered = apply_hard_constraints(geo_filtered)

    # Stage 3: Score and rank
    return argmax(hc_filtered, key=lambda r: PS_Prod(r))
```

Examples

Customer Geography	Eligible Regions	Ineligible (HC-8)
US	East US, West US, Central US, …	West Europe, UK South, …
Europe	West Europe, North Europe, UK South, …	East US, Australia East, …
Middle East	UAE North, Saudi Arabia Central	East US, Switzerland North*

Switzerland North becomes eligible only via HC-10 Cross-Geo Extension Path for DR, not for Prod — and that path is inactive* for the Middle East while ME DR is `DR_NOT_OFFERED` (DR-014, pending DEC-001 legal approval).

Evidence Tag

`[Undocumented — architectural judgement]`

HC-9: STANDARD_REGION_ONLY [NEW]

Category: Region Class
Source: Production Readiness Review §27
Status: New gate for production-ready region classification

Definition

```
Automated placement, scoring, recommendation, and all environment assignments
(Prod, CVAL, DR) must use Standard Capacity Regions only.
```

Rationale

Restricted Capacity Regions (sovereign clouds, government regions, regions with SKU limitations) are excluded from automated placement. Their exclusion is enforced at **Stage 1 of the eligibility decision tree**, not as a scoring penalty.

Exception deployments (customer explicitly requests a Restricted region) proceed via the **Scenario 2 exception path** only, requiring operator approval.

Region Classification

Standard Capacity Regions:
- Commercial Azure public cloud regions with full SKU availability
- No sovereign cloud restrictions
- No special approval required for CRG creation
- Examples: East US, West Europe, Australia East, Japan East, Brazil South

Restricted Capacity Regions:
- Azure Government (US DoD, US Gov)
- Azure China (21Vianet)
- Regions with limited SKU availability (flagged in Azure metadata)
- Regions requiring special tenant approval (sovereign cloud onboarding)

Enforcement Stage

Stage 1 — Pre-filtering (before candidate set formation)

Validation Rules (VR)

- **VR-5:** All automated placement paths (Scenario 1 geography-based, Prod derivation, NonProd/DR selection) must use Standard Capacity Regions only
- **VR-6:** CVAL and DR must not use Restricted Capacity Regions under any condition, including exception deployments

Implementation

```
def get_eligible_regions_stage1(geography, customer_request):
    all_regions = get_all_azure_regions()

    # HC-9: Standard regions only (Stage 1 pre-filter)
    standard_regions = [r for r in all_regions if r.region_class == "Standard"]

    # If customer explicitly requests a Restricted region → Scenario 2 exception path
    if customer_request.region_override and customer_request.region_override.re-
gion_class == "Restricted":
        return handle_exception_deployment(customer_request)

    # HC-8: Geography containment
    geo_filtered = [r for r in standard_regions if r.geography == geography]

    return geo_filtered
```

Exception Deployment Workflow (Scenario 2)

Trigger: Customer explicitly requests a Restricted Capacity Region

Workflow:
1. Operator receives request via exception deployment queue
2. Validate customer has required tenant permissions (sovereign cloud registration)
3. Validate SKU availability in target Restricted region
4. Manual approval gate (ops team review)
5. If approved: Create Prod CRG in Restricted region; **CVAL and DR must still use Standard regions** (VR-6)
6. Record exception in `OperationRecord` with approval metadata

Evidence Tag

`[Undocumented — architectural judgement]`

HC-10: CROSS_GEO_EXTENSION_PATH_APPROVED [NEW]

Category: Cross-Geo DR
Source: Production Readiness Review §27
Status: New governance control for cross-geography DR assignments

Definition

```
Any DR assignment to a region outside the customer's chosen geography must match
an explicitly enumerated Cross-Geo Extension path in the active PlacementPolicy.
```

Rationale

Cross-geography DR introduces data residency, compliance, and latency considerations that require explicit approval. The engine does not autonomously select cross-geo DR regions; it only accepts pre-approved extension paths defined in the active `PlacementPolicy`.

Enforcement

DR assignments to Standard Capacity Regions in a **different geography** are rejected if no approved extension path exists for the source geography.

Approved Cross-Geo Extension Paths (Example)

Source Geography	Approved DR Geographies	Primary DR Region	Rationale
Middle East	Europe	Switzerland North **(inactive — gated by DEC-001)**	Middle East has only 2 regions (UAE North, Saudi Arabia); a 3-region model would need cross-geo DR, but ME DR is currently `DR_NOT_OFFERED` (data-sovereignty/residency, DR-014). Path stays inactive until Legal approves DR (DEC-001).
Australia	Asia Pacific	Singapore	Regional pair fallback
India	Asia Pacific	Singapore	Regional pair fallback

Middle East Three-Region Placement (Conditional Example — gated by DEC-001)

> **This example applies ONLY if Legal approves Middle East DR (DEC-001).** As it stands, the current legal position is `DR_NOT_OFFERED` for the Middle East (data-sovereignty/residency; see the ⚠️ note at the top of this document and DR-014). Under the default legal state the placement is: **Prod** in-geo, **CVAL/NonProd** in-geo, and `dr_region = NOT_OFFERED` — steps 3 below are **not executed** and no cross-geo DR is assigned. The steps below describe the pre-configured behaviour that becomes live **only after** DEC-001 records an approval.

Inputs:
- Customer geography: Middle East
- Available Middle East Standard regions: UAE North, Saudi Arabia Central (2 only)
- **Middle East DR policy flag (DR-014):** default `DR_NOT_OFFERED = true` (pending DEC-001)

Default placement (current legal position — `DR_NOT_OFFERED`):
1. **Prod:** `argmax(PS_Prod)` over {UAE North, Saudi Arabia Central} → e.g., UAE North
2. **CVAL/NonProd:** Remaining Middle East region → Saudi Arabia Central
3. **DR:** `NOT_OFFERED` — seed records no DR region; engine does not assign Switzerland North or any other cross-geo region. Middle East production may still exist without DR.

Conditional placement (only if DEC-001 approves ME DR — sets `DR_NOT_OFFERED = false`):
1. **Prod:** `argmax(PS_Prod)` over {UAE North, Saudi Arabia Central} → e.g., UAE North
2. **CVAL/NonProd:** Remaining Middle East region → Saudi Arabia Central
3. **DR:** Cross-Geo Extension to **Switzerland North** (pre-approved path, now activated)

Validation (conditional path only):
- Switzerland North must pass **HC-1 through HC-10** including DR coverage floor (HC-6)

- If Switzerland North fails any HC, the placement is **rejected with an ops alert**
- The engine does **not** silently select any alternative outside the approved extension paths

Validation Rules (VR)

- **VR-8 (default):** For a Middle East geography with `DR_NOT_OFFERED = true` (current legal position, DEC-001), the seed **must** record `dr_region = NOT_OFFERED`; the engine must **not** auto-assign Switzerland North or any cross-geo DR.
- **VR-8a (conditional):** Only if DEC-001 approval has cleared the `DR_NOT_OFFERED` flag, the Middle East DR region must be Switzerland North via the approved Cross-Geo Extension path.
- **VR-11:** Cross-Geo Extension DR path must be explicitly approved in active `PlacementPolicy` **and** (for the Middle East) gated by a recorded DEC-001 approval.

Implementation

```python
def resolve_dr_region(customer_geography, candidate_dr_region):
    policy = get_active_placement_policy()

    # DR-014 / DEC-001: honour the per-geography DR_NOT_OFFERED legal flag FIRST.
    # For the Middle East this defaults to True (current legal position) until Legal
    # records a DEC-001 approval that clears it.
    if policy.dr_not_offered.get(customer_geography, False):
        return "NOT_OFFERED"  # seed records no DR; no cross-geo substitution (VR-6)

    return candidate_dr_region


def validate_HC10_cross_geo_extension(customer_geography, dr_region):
    policy = get_active_placement_policy()

    # DR_NOT_OFFERED short-circuits HC-10: no DR region to validate.
    if dr_region == "NOT_OFFERED":
        return PASS  # legal no-DR outcome (DR-014, DEC-001)

    # Check if DR region is in same geography as customer
    if dr_region.geography == customer_geography:
        return PASS  # Same-geo DR, HC-10 does not apply

    # Cross-geo DR → validate against approved extension paths
    approved_paths = policy.cross_geo_extension_paths.get(customer_geography, [])

    if dr_region not in approved_paths:
        return REJECT, f"HC-10 CROSS_GEO_EXTENSION_PATH_APPROVED: DR region
{dr_region.name} not in approved extension paths for {customer_geography}"

    # DEC-001 gate for the Middle East: even a configured extension path stays
    # inactive until a recorded legal approval clears DR_NOT_OFFERED.
    if customer_geography == "Middle East" and policy.dr_not_offered.get("Middle
East", True):
        return REJECT, "HC-10 DEC-001 PENDING: Middle East DR is DR_NOT_OFFERED
pending legal approval; Switzerland North extension is inactive"

    return PASS
```

PlacementPolicy Configuration (Example)

```
{
  "policy_version": "2026-08-Q3",
  "dr_not_offered": {
    "Middle East": true
  },
  "dr_not_offered_rationale": {
    "Middle East": "Legal-owned programme (DEC-001, under review). Data-sovereignty/
residency laws for a largely government/medical customer base mean cross-border DR
cannot meet residency requirements; DR is currently NOT offered (baseline §2/§5.2/§6,
DR-014). Production may still exist without DR. Set to false ONLY when Legal records a
DEC-001 approval."
  },
  "cross_geo_extension_paths": {
    "Middle East": {
      "approved_dr_regions": ["Switzerland North"],
      "activation_gated_by": "DEC-001 (dr_not_offered['Middle East'] must be false)",
      "rationale": "Pre-configured, CONDITIONAL path. Middle East has only 2 Standard
regions, so a 3-region model would need cross-geo DR — but this path is inactive while
DR_NOT_OFFERED is true."
    },
    "Australia": {
      "approved_dr_regions": ["Singapore", "Japan East"],
      "rationale": "Regional pair fallback for Australia East / Australia Southeast"
    }
  }
}
```

Evidence Tag

`[Undocumented — architectural judgement]`

Part 2 — Hard Constraint Summary Table

HC	Name	Type	Enforcement Stage	Primary Impact	Updated in Pass
HC-1	REGION_SEPARATION	Isolation	Stage 2	Prod isolated from Non-Prod/DR; NonProd=DR co-location allowed	Pass 2 (D8)
HC-2	CAPACITY_FLOOR	Capacity	Stage 2	Minimum CR headroom required	Stable
HC-3	QUOTA_FLOOR	Quota	Stage 2	Minimum quota headroom per environment type (Prod/NonProd/DR)	Pass 2 (QG-3)
HC-4	DR_SEPARATION_CLASS	Resilience	Stage 2	DR region must have HIGH separation class from Prod	Stable
HC-5	ZONE_AVAILABILITY	Resilience	Stage 2	Minimum 2 availability zones required	Stable
HC-6	DR_COVERAGE_FLOOR	DR Capacity	Stage 2	Combined DR+NonProd capacity must absorb DR demand	Pass 2 (D8)
HC-7	DR_FLOOR_INTEGRITY	Quota Protection	Stage 2	NonProd placement cannot encroach on DR quota floor	Pass 2 (QG-4)
HC-8	GEOGRAPHY_CONTAINMENT	Geography	Stage 1	Prod region must be within customer’s	Implied

HC	Name	Type	Enforcement Stage	Primary Impact	Updated in Pass
				chosen geography (Scenario 1)	
HC-9	STANDARD_REGION_ONLY	Region Class	Stage 1	Only Standard Capacity Regions eligible for automated placement	PRR §27
HC-10	CROSS_GEO_EXTENSION_PATH_APPROVED	Cross-Geo DR	Stage 2	Cross-geography DR requires explicit approval path	PRR §27

Part 3 — Enforcement Pipeline

Stage 1 — Pre-filtering (Eligibility Decision Tree)

Applied before candidate set formation:

```
All Azure Regions
    ↓
[HC-9: STANDARD_REGION_ONLY filter]
    ↓
Standard Capacity Regions
    ↓
[HC-8: GEOGRAPHY_CONTAINMENT filter] (if Scenario 1)
    ↓
Standard Regions in Customer Geography
    ↓
Candidate Set for Stage 2
```

Stage 2 — Hard Constraint Gate

Applied to each candidate region before scoring:

```
def apply_hard_constraints_stage2(region, environment_type, customer_context):
    """
    Returns: (PASS, None) or (REJECT, reason)
    """
    # HC-1: Region separation
    if not validate_HC1_region_separation(region, environment_type, customer_context):
        return REJECT, "HC-1 REGION_SEPARATION"

    # HC-2: Capacity floor
    if not validate_HC2_capacity_floor(region, environment_type):
        return REJECT, "HC-2 CAPACITY_FLOOR"

    # HC-3: Quota floor
    if not validate_HC3_quota_floor(region, environment_type, customer_context):
        return REJECT, "HC-3 QUOTA_FLOOR"

    # HC-4: DR separation class (DR only)
    if environment_type == "DR" and not validate_HC4_dr_separation_class(region, customer_context.prod_region):
        return REJECT, "HC-4 DR_SEPARATION_CLASS"

    # HC-5: Zone availability
    if not validate_HC5_zone_availability(region):
        return REJECT, "HC-5 ZONE_AVAILABILITY"

    # HC-6: DR coverage floor (DR only)
    if environment_type == "DR" and not validate_HC6_dr_coverage_floor(region, customer_context):
        return REJECT, "HC-6 DR_COVERAGE_FLOOR"

    # HC-7: DR floor integrity (NonProd only)
    if environment_type == "NonProd" and not validate_HC7_dr_floor_integrity(region, customer_context):
        return REJECT, "HC-7 DR_FLOOR_INTEGRITY"

    # HC-10: Cross-geo extension path (cross-geo DR only)
    if environment_type == "DR" and region.geography != customer_context.geography:
        if not validate_HC10_cross_geo_extension(customer_context.geography, region):
            return REJECT, "HC-10 CROSS_GEO_EXTENSION_PATH_APPROVED"

    return PASS, None
```

Stage 3 — Scoring Pipeline

Regions surviving HC-1..HC-10 enter scoring:

- **Prod:** Ranked by `PS_Prod(r)` = argmax over survivors
- **NonProd:** Ranked by `PS_NonProd(r)` = argmax over survivors (excluding Prod region)
- **DR:** Ranked by `PS_DR(r)` = argmax over survivors (excluding Prod region; may include NonProd region per HC-1)

Part 4 — Validation Rules (VR) Cross-Reference

The Production Readiness Review defines **11 Validation Rules (VR-1..VR-11)** that enforce hard constraints:

VR	Description	Enforced HCs
VR-1	Prod and DR must not share a region	HC-1
VR-2	CVAL and Prod must not share a region	HC-1
VR-3	CVAL and DR may share a region only under approved policy	HC-1 (D8 update)
VR-4	Derived Prod region must be within customer's chosen geography	HC-8
VR-5	All automated placement paths must use Standard Capacity Regions only	HC-9
VR-6	CVAL and DR must not use Restricted Capacity Regions under any condition	HC-9
VR-7	Standard region passes HC-1 through HC-10 → if all excluded, exhaustion error	HC-1..10
VR-8	For Middle East geography, DR defaults to `NOT_OFFERED` (DR-014, DEC-001 legal position); Switzerland North via approved cross-geo path applies **only after** a recorded DEC-001 approval clears `DR_NOT_OFFERED`	HC-10
VR-9	Only on the DEC-001-approved conditional path: if Switzerland North fails HC-1..HC-10, block placement with ops alert	HC-1..10
VR-10	Restricted region requested by customer → Scenario 2 exception path only	HC-9
VR-11		HC-10

VR	Description	Enforced HCs
	Cross-Geo Extension DR path must be explicitly approved in active PlacementPolicy	

Part 5 — POC Blockers & Testing

Critical POC Blockers

ID	Issue	Blocked HCs	POC Gate	Resolution
B-1	Azure Quota Groups functionality in target tenant/region	HC-3, HC-7	POC-30	GA availability check; if 404 → escalate to Azure Support
B-2	Quota pool release behavior on CR reduction (Tier 2/3 quota-neutral claim)	HC-7	POC-31	Measure release latency; confirm < 5 min for Tier RTO

POC Test Coverage

POC-01 through POC-10: Hard constraint validation (HC-1..HC-10)
POC-11: Middle East cross-geo DR (HC-10) — **blocked pending DEC-001**; ME DR is `DR_NOT_OFFERED` today, so this POC runs only if/when Legal approves ME DR
POC-30: Quota Groups API integration (HC-3, HC-7)
POC-31: Quota pool release latency (HC-7 emergency transfer dependency)

Part 6 — Configuration & Policy

PlacementPolicy Schema (Hard Constraint Section)

```json
{
  "policy_version": "2026-08-Q3",
  "hard_constraints": {
    "capacity_floor_multiplier": 2,
    "dr_ratio_min": 0.30,
    "dr_ratio_max": 0.40,
    "dr_ratio_target": 0.35,
    "allow_single_zone_dev": false,
    "cross_geo_extension_paths": {
      "Middle East": {
        "approved_dr_regions": ["Switzerland North"],
        "rationale":
"Middle East has only 2 Standard regions; 3-region model requires cross-geo DR"
      }
    },
    "separation_class_overrides": {}
  }
}
```

Policy Constants

Constant	Default Value	Configurable?	Affected HCs
capa-city_floor_multiplier	2	✅ Yes (per geo-graphy/SKU)	HC-2
dr_ratio_min	0.30	✅ Yes	HC-6, HC-7
dr_ratio_max	0.40	✅ Yes	HC-6, HC-7
dr_ratio_target	0.35	✅ Yes	Scoring (not HC)
min_zone_count	2	✅ Yes	HC-5
al-low_single_zone_dev	false	✅ Yes	HC-5
cross_geo_extension_paths	{}	✅ Yes	HC-10

Part 7 — Implementation Checklist

Engineering Backlog Cross-Reference

HC	Primary Epic	Stories	Tasks
HC-1..HC-8	EPIC-07 (Regional Snapshot), EPIC-08 (Placement)	S0702, S0703, S0704, S0705	T070201
HC-9, HC-10	EPIC-07 (Regional Snapshot), EPIC-08 (Placement)	S0702	T070202
HC-3, HC-7 (Quota Groups)	EPIC-03 (Quota Management)	S0310, S0311, S0314	T031001..T031006
HC-6 (DR Coverage)	EPIC-06 (DR Management), EPIC-08 (Placement)	S0601, S0704	T060101, T070401

Pre-Deployment Validation

- [ ] HC-1..HC-10 validation functions implemented (unit tested)
- [ ] Stage 1 pre-filtering logic (HC-8, HC-9) validated
- [ ] Stage 2 hard constraint gate logic (HC-1..HC-7, HC-10) validated
- [ ] RegionalSnapshot schema extended with HC-6 fields (nonprod_crg_effective_free, dr_crg_free_slots)
- [ ] QuotaGroup integration (HC-3, HC-7) validated via POC-30
- [ ] Cross-Geo Extension paths configured in PlacementPolicy (Middle East path present but **inactive** — `dr_not_offered["Middle East"] = true` pending DEC-001)
- [ ] Middle East default placement validated: `dr_region = NOT_OFFERED` (DR-014, current legal position)
- [ ] Middle East conditional three-region placement (Switzerland North, HC-10) validated **only after** DEC-001 legal approval
- [ ] Exception deployment workflow (HC-9 Restricted region override) tested
- [ ] HC rejection reasons logged to OperationRecord for audit
- [ ] Validation Rules VR-1..VR-11 traced to HC enforcement points

Appendix A — Decision Log Cross-Reference

Decision	HCs Affected	Change
D8	HC-1, HC-6	Removed `DR_region ≠ Non-Prod_region` constraint; added HC-6 to operationalise co-location
D1, D2	HC-3, HC-7	Introduced Two-Group Quota Architecture; HC-3 now reads from Quota Groups; HC-7 enforces DR floor
D9	HC-9, HC-10	Introduced Standard/Restricted region classification; added HC-9 and HC-10 governance gates

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Appendix B — Evidence Tags

All hard constraints carry one of four evidence tags:

- **[Documented]** — Derived from Azure platform documentation (e.g., Quota Groups API)
- **[Derived]** — Logical consequence of a documented constraint or design decision
- **[Decided]** — Explicit design choice (traceable to Decision Log D1–D9)
- **[Undocumented — architectural judgement]** — Azure Well-Architected Framework best practice, not explicitly documented by Microsoft

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Document Status: Production Ready
Next Review: After POC-30 and POC-31 completion (Quota Groups GA validation)